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A platform for creating computational exposure models: personalized dosimetry with Holmium-166

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Farias, W. A. W. A.[1], Amaral, R. S.[1], Vieira, J. W.[2], Andrade, P. H. A .[3], Bezerra, L. R. D. A .[3], Lima, F. R. A .[4], Melo, S. B.[5], Santos, R. C. M .[1].

[1]DEN/UFPE – Recife, Brasil; [2]IFPE – Recife, Brasil; [3]IFPE – Vitória de Santo Antão, Brasil; [4]CRCN-NE – Recife, Brasil; [5]CIN/UFPE – Recife, Brasil.

***Correspondence to: whoody.alem@ufpe.br** **@whoody_alem**

INTRODUCTION

The PHAntoms MAnufacturing (PHAMA) add-on was developed to extend Blender with specific tools for creating, editing, and exporting BREP phantoms, as well as generating radiation source algorithms [1]. Integrated with other add-ons and the Digital Imaging Processing (DIP) software [2], PHAMA optimizes computational dosimetry workflows using EGSnrc and other compatible Monte Carlo codes. This work presents the application of PHAMA in the construction of a computational exposure model (CEM) with the Personalized Anatomy Model (PAM) phantom and a Holmium-166 (Ho-166) source coupled to EGSnrc.

MATERIALS AND METHODS

• Blender v. 2.83: Open-source 3D environment for polygonal mesh modeling;

• PHAMA v.1.1.0 (in-house): Blender add-on developed for creating and editing anthropomorphic phantoms and modeling the Ho- 166 source;

• OrtogOnBlender v. 2021.12.25: Add-on used to convert DICOM images to meshes;

• Digital Image Processing (DIP): Software for voxelizing the PAM, creating the probability distribution function (PDF), cumulative distribution function (CDF), and generating input files for the MCE;

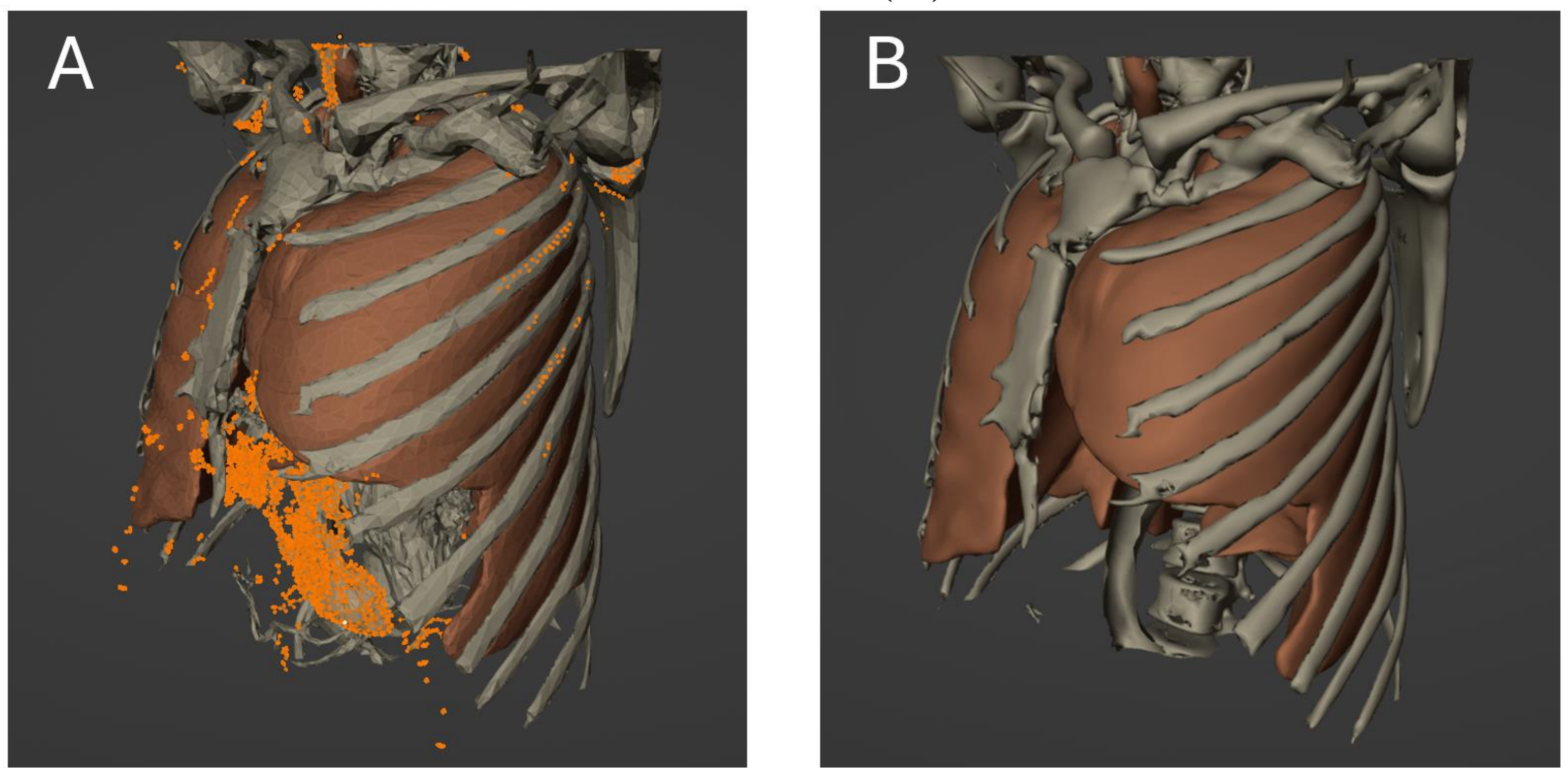
• EGSnrc: Monte Carlo code for radiation transport and interaction, with analysis of deposition in regions of interest.

The PAM was constructed based on computed tomography images of the chest and abdomen of a male patient from a public database. To this end, the DICOM images were converted into polygonal meshes using OrtogOnBlender. PHAMA was used for both structural adjustments of the phantom (removing loose vertices and closing gaps) and for modeling the Ho-166 source, defining the boundary surfaces and generating the output file containing the relative heights corresponding to the emission probability β^- as a function of energy. The final PAM model was exported in .obj format and subsequently processed in DIP, which converted the PAM to voxels (.SID) and generated the EGSnrc input file for the source.

RESULTS

The integration between Blender, OrtogOnBlender, and PHAMA allowed the conversion of DICOM images into polygonal meshes. PHAMA was used for mesh editing, as shown in Figure 1, and spectral modeling, while DIP converted the PAM into voxels and prepared the files for EGSnrc.

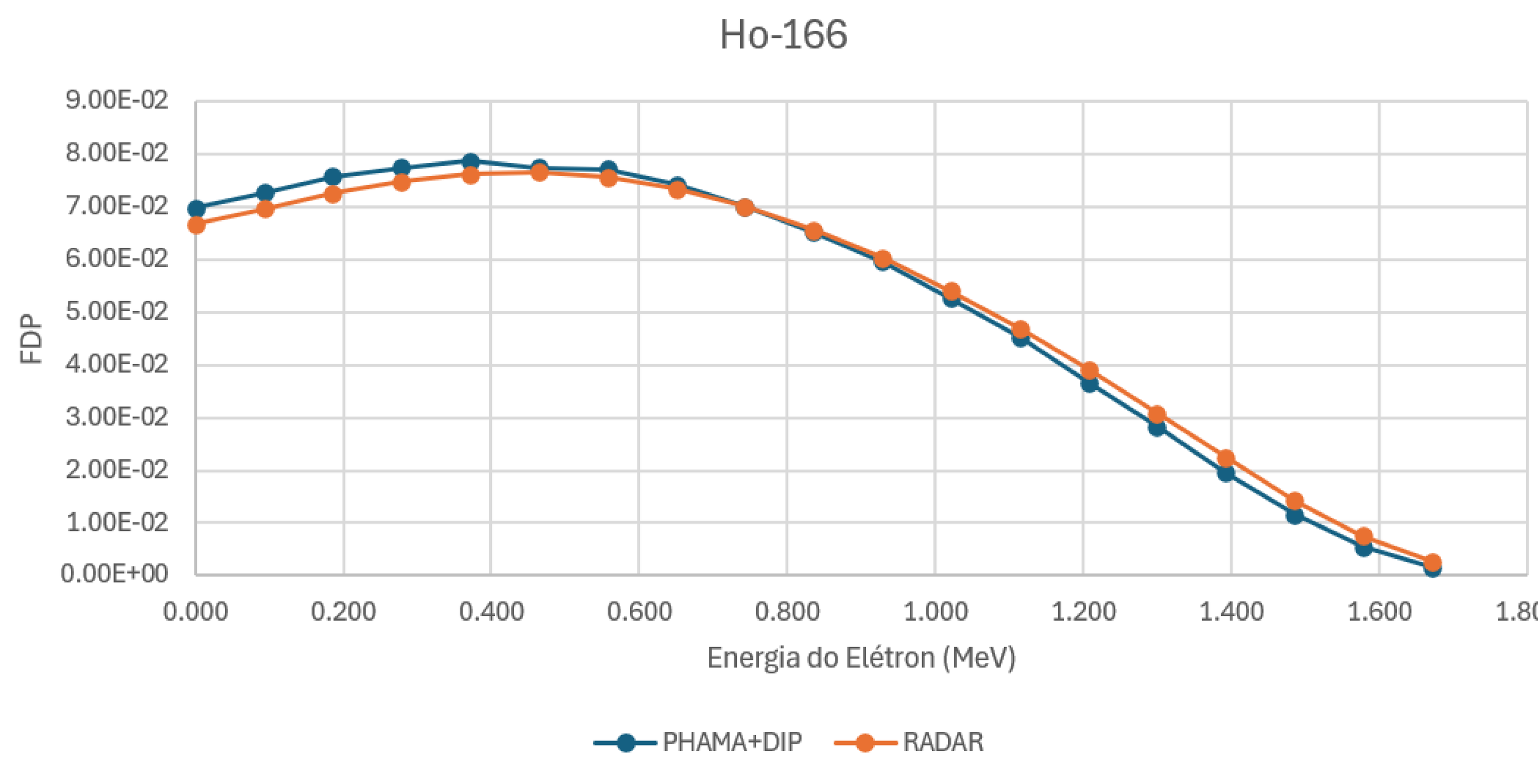
Figure 1. PAM. Selection of vertices for deletion (A), result of modifications (B).



Source: the authors, 2025.

The analysis of the PDF obtained by PHAMA/DIP shows high consistency with the reference data from the Radiation Dose Assessment Resource (RADAR) as shown in Graph1.

Graph 1. PDF for Ho-166.



Source: the authros, 2025.

CONCLUSIONS

PHAMA, in combination with DIP, proved to be an effective tool for creating, editing, and exporting anthropomorphic phantoms and for modeling radioactive sources; the generated data provide a solid foundation for future simulations with a custom phantom and a Ho-166 source.

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